

COOL-Edit: Preventing Neuron Overheating for Serial Lifelong Knowledge Editing

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Abstract—Large language models (LLMs) acquire vast factual knowledge during pre-training, but this knowledge quickly becomes stale as the world changes. Knowledge editing has emerged as an efficient alternative to costly fine-tuning for updating outdated or incorrect facts without retraining. However, existing methods are designed for low-frequency, one-shot updates and degrade sharply under serial lifelong editing, where the same fact must be revised repeatedly at high frequency. We uncover a key failure mode: when the same subject-relation pair is edited over and over, mainstream editors such as AlphaEdit repeatedly overwrite an identical, localized subset of neurons. The accumulated perturbations on these neurons, which we term **neuron overheating**, cause catastrophic interference and ultimately model collapse. To address this, we propose COOL-Edit (Cold-neuron Optimized Overheating-Limited Knowledge Editing), a heat-guided editing method that maintains a cumulative heat value for each neuron and reduces the probability of re-updating hotter neurons, thereby spreading successive edits across the model and preventing any single neuron from overheating. To evaluate high-frequency editing, we construct CounterFact-freq, a 2,000-step serial lifelong editing benchmark derived from CounterFact. Experiments on LLaMA-3-8B show that COOL-Edit consistently outperforms strong baselines, including AlphaEdit and NMKE, in Efficacy, Generalization, and Specificity, while better preserving the model’s general capabilities on MMLU. Source code and datasets are available in this GitHub repository: <https://github.com/PrayerVoid/COOL-Edit>.

Index Terms—knowledge editing, serial lifelong editing, neuron selection, large language models

I. INTRODUCTION

Large language models [1]–[4] (LLMs) have demonstrated remarkable capabilities in storing extensive knowledge during pre-training and recalling it during inference. However, as the real world continuously changes, LLMs frequently suffer from hallucination phenomena [5], generating incorrect or outdated information [6]. While fine-tuning the model with updated knowledge offers a direct solution, it is frequently hampered by excessive computational costs and the potential for catastrophic forgetting [7], [8]. Consequently, knowledge editing has emerged as an effective paradigm.

Existing knowledge editing methodologies can be broadly divided into two categories: parameter-preserving approach (e.g., retrieval-augmented generation [9] and in-context learning [10]) and parameter-modifying approach. Within the lat-

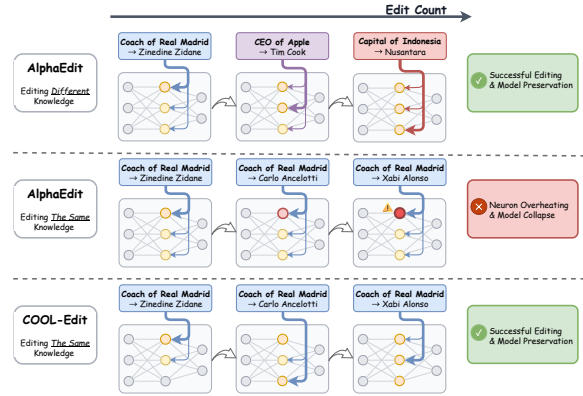


Fig. 1. Illustration of the neuron overheating phenomenon and how COOL-Edit addresses it.

ter, a highly representative strategy is the “Locate-then-Edit” method (e.g., ROME [11] and AlphaEdit [12]), which explicitly pinpoints the exact components in LLMs responsible for specific knowledge and applies constrained updates to those precise parameter subspaces.

However, current methodologies are primarily designed and evaluated for low-frequency knowledge editing—assuming different facts are edited sporadically. As illustrated in Fig. 1, when AlphaEdit updates different knowledge, it alters disjoint neural subsets, resulting in successful edits and model preservation. But in reality, numerous facts demand recurrent, high-frequency modifications [13], [14]. When tasked with such sequential editing on the same knowledge, AlphaEdit exhibit severe performance degradation.

Our empirical analysis explains this degradation. In a simplified repeated-editing experiment, we observe that AlphaEdit concentrates its updates on a subset of neurons, resulting in substantial parameter deformation of these neurons — a phenomenon we term **Neuron Overheating**. These overheated neurons can trigger model collapse and degrade editing performance, as shown in the second line of Fig. 1. We then further validate this claim through a neuron-restoration experiment.

To address this challenge, we introduce **Cold-neuron Optimized Overheating-Limited Knowledge Editing** (COOL-

Edit), a fine-grained approach designed to mitigate the negative effects of high-frequency knowledge updates. Unlike existing methods that simply modify the related neurons for each knowledge, COOL-Edit prioritizes neurons with low historical update magnitudes, thereby avoiding redundant interventions on previously edited neural subsets.

Our paper presents the following contributions:

- **Discovery of the Neuron Overheating Phenomenon:** We uncover that sequentially updating the target object from o to $o_1^*, o_2^*, \dots, o_n^*$ for the same (s, r) pair disproportionately concentrates on a localized subset of neurons, causing substantial parameter distortion (overheating) within these neurons and subsequent model collapse.
- **A Hotness-Aware Neuron Selection Approach:** Driven by this empirical observation, we propose COOL-Edit, a novel knowledge editing methodology with a heat-guided neuron selection mechanism.
- **New Benchmark Construction:** We construct a novel dataset CounterFact-freq based on the CounterFact [11], explicitly tailored for high-frequency knowledge updates.

The paper is organized as follows. Section II briefly reviews related work on knowledge editing. Section III formulates the serial lifelong knowledge editing problem and presents an empirical analysis of repeated editing, establishing three converging observations on update concentration, parameter deformation, and their strong negative correlation, supported by a neuron-restoration experiment that confirms overheated neurons as the main cause of model collapse. Section IV introduces the COOL-Edit framework, detailing its heat-aware neuron selection mechanism via softmax sampling and Bernoulli masking, the complete editing algorithm, a preliminary validation on the 200-edit task, and a complexity analysis. Section V presents the CounterFact-freq benchmark, the main experimental results on the 2,000-step serial lifelong editing task, and an analysis of optimization-step dynamics under repeated editing. Section VI concludes the paper, and Section VII outlines future research directions.

II. RELATED WORK

A. Knowledge Editing Methods

Knowledge editing methods can be divided into two core strategies based on whether the original parameters of LLMs are preserved. The parameter-preserving approaches realize knowledge editing by storing edited knowledge in external memory without modifying LLM parameters, with representative methods including SERAC [16], MeLLO [15], and IKE [10]. In comparison, parameter-modifying approaches directly adjust internal model parameters for knowledge update, which consist of two mainstream branches: hypernetwork-based methods [17] and the prevalent “Locate-then-Edit” paradigm.

As a mainstream solution in parameter-modifying editing methods, the “Locate-then-Edit” paradigm is built on the core perspective that Feed-Forward Networks (FFNs) in Transformers act as key-value memories [19]. Specifically,

the foundational work ROME [11] pioneered this paradigm by directly modifying FFN weights in a key layer to inject factual knowledge. Then, MEMIT [20] extended ROME by updating multiple layers and enabling batch editing. To further facilitate continual knowledge editing, Fang et al. introduced AlphaEdit [12], one of the dominant approaches, which achieves more stable knowledge editing by projecting parameter perturbations onto the null space of the key matrices. Nowadays, numerous subsequent studies have attempted to improve AlphaEdit and achieved promising results, exemplified by NMKE [21] and AlphaEdit+ [18]. Our work is also built upon AlphaEdit.

B. Knowledge Editing Tasks

Knowledge editing has evolved from static, single-round updates to dynamic, lifelong maintenance of LLMs. Early research primarily focused on single-round or batch editing tasks [11], [20], where a model is updated with a discrete fact or a static set of facts simultaneously. However, in real-world applications, knowledge is inherently fluid and evolves continuously [6]. This necessity has spurred the development of lifelong editing [23], where models must incorporate a persistent, streaming flow of updates over time.

While classic lifelong editing focuses on sequential updates for unique and distinct facts, real-world scenarios often require updating the exact same concept multiple times as facts change. To address this, recent work introduced Serial Lifelong Knowledge Editing (sLKE) [14], a more challenging and realistic setting that focuses on sequentially and repeatedly updating answers to the same question over time. Despite its practical value, sLKE remains a relatively under-studied task, and existing knowledge-editing approaches have not been thoroughly evaluated in this setting.

III. EMPIRICAL OBSERVATIONS

In this section, we first formulate the problem of knowledge editing in the context of serial lifelong knowledge editing in Section III-A, and then present our empirical analysis of repeated editing in Section III-B.

A. Problem Statement

Knowledge editing aims to efficiently update a fact triple from (s, r, o) to (s, r, o^*) , where s , r , and o denote the subject, relation, and object of the fact, respectively. For example, given the question “Who is the president of the United States?”, a model may answer “Joseph Robinette Biden” because of outdated knowledge. The corresponding editing request is $(s = \text{United States}, r = \text{president}, o = \text{Joseph Robinette Biden}, o^* = \text{Donald John Trump})$; after editing, the model should answer “Donald John Trump” rather than “Joseph Robinette Biden”. The goal of editing is to make the model output the new object o^* for the prompt formed by s and r (and its paraphrases), while leaving the remaining knowledge unchanged.

Building on this, our study targets *serial lifelong knowledge editing* (sLKE) [14]: the model performs a sequence

of edits on a stream of editing requests, covering multiple pieces of knowledge, where the same piece of knowledge may be edited multiple times. In real-world scenarios, some knowledge is updated frequently (e.g., the head of an organization). We demonstrate that the mainstream method AlphaEdit [12] causes the model to collapse rapidly under such high-frequency updates of the same knowledge, analyze the underlying reason, and design a novel editing method, COOL-Edit, accordingly. To evaluate the effectiveness of our method, we construct a dataset CounterFact-freq based on CounterFact [11] that contains knowledge with various update frequencies, and evaluate different methods on this task.

B. Empirical Observations on Repeated Editing

To study the behavior of AlphaEdit under high-frequency editing of the same knowledge, we design a simplified *repeated-editing* experiment. Using AlphaEdit as the base editor, we repeatedly edit the fact “Danielle Darrieux’s mother tongue is French” 200 times on LLaMA-3-8B-Instruct [2], alternating the target value between English and Spanish, and compare against *sequential editing*, where 200 different facts are each edited once in sequence. We mainly evaluate the editing results with three metrics: Efficacy, Generalization, and Specificity [11]. To better fit this task setting, when computing the metrics we require the probability of the correct object o to be greater than that of all the remaining objects; see Appendix B.

Fig. 2 shows the evolution of the three metrics over the course of editing (cumulative means). Repeated editing degrades all three metrics, with Specificity degrading most severely. This indicates that AlphaEdit performs poorly when the same knowledge is updated at high frequency.

To analyze the reason behind this phenomenon, we further analyze the edited model parameters. For each edited layer and each neuron, we compute the sum of the norms of the update vectors acting on that neuron over the 200 edits (*cumulative update magnitude*), and after editing we compute the cosine similarity between the final parameter vector of each neuron and its original one (*final cosine similarity*). We use these quantities to analyze how the model parameters change during editing. Here we take layer 6 as an example; the other layers behave similarly, and the complete results are deferred to Appendix C.

Observation 1 (Update concentration). Fig. 3(a) and (b) give two pieces of evidence for update concentration on layer 6. (a) The Lorenz curve of cumulative update magnitudes: the curve of repeated editing deviates clearly from the line of equality ($y = x$) and lies well above that of sequential editing; the Gini coefficient of repeated editing is about 0.240, while that of sequential editing is about 0.155. (b) The average pairwise cosine similarity of the update vectors across edits: 0.564 for repeated editing, clearly higher than 0.463 for sequential editing, indicating that the update directions are more consistent under repeated editing. Together, these two pieces of evidence show that, under repeated editing, updates accumulate more heavily on a subset of neurons.

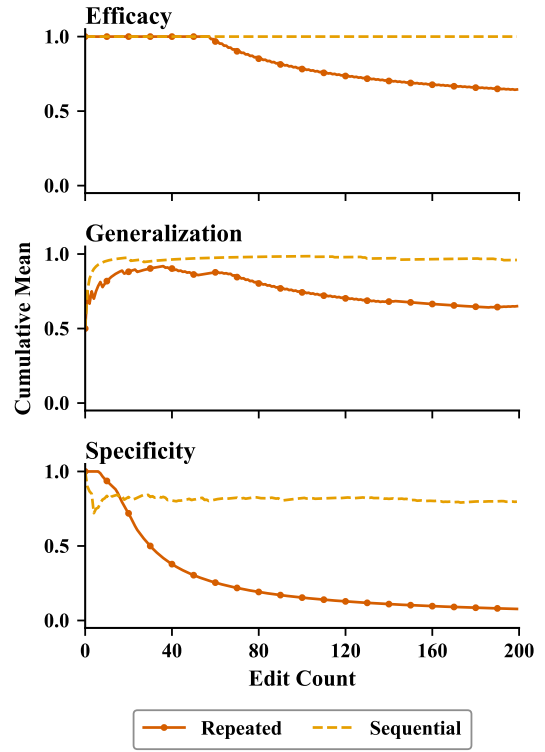


Fig. 2. Cumulative means of the three metrics over 200 edits for repeated editing (Repeated) vs. sequential editing (Sequential).

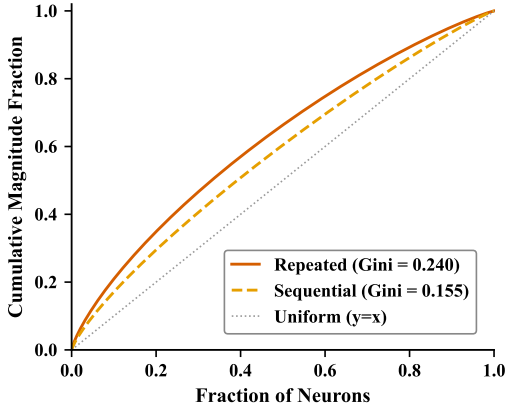
Observation 2 (Parameter deformation). Fig. 3(c) shows that, after 200 edits, the layer-6 mean per-neuron cosine similarity of repeated editing (0.653, median 0.671) is far lower than that of sequential editing (0.997, median 0.997); the neuron parameters have been severely deformed.

Observation 3 (Strong negative correlation). Fig. 3(d) plots the cumulative update magnitude against the final cosine similarity neuron by neuron for AlphaEdit on layer 6: the two are tightly negatively correlated, with Spearman $\rho = -0.990$. That is, the neurons updated most frequently are exactly the ones deformed most severely.

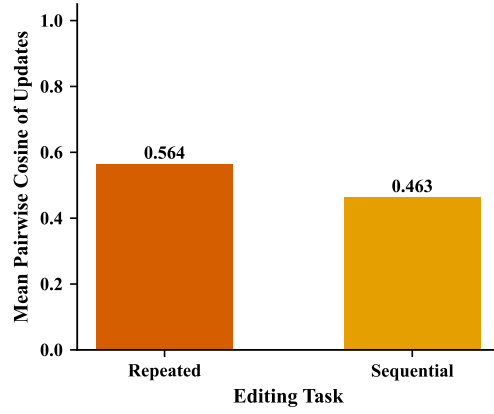
Combining these three observations, we naturally hypothesize that, compared with ordinary sequential editing, the abnormal performance degradation under repeated editing is caused by updates being concentrated on a small number of neurons, which become severely deformed and lead to model collapse. We term this phenomenon **neuron overheating**.

To further verify this hypothesis, we conduct a neuron-restoration experiment (Fig. 4): starting from the AlphaEdit model after 200 repeated edits, we restore neurons layer by layer to their original parameters, either in descending order of cumulative update magnitude (top- k) or randomly, and measure Specificity as a function of the fraction of restored neurons. Top- k restoration brings Specificity back to 1.0 when 50% of the neurons are restored, whereas random restoration requires 80%. This shows that the overheated neurons are indeed a major cause of the performance degradation.

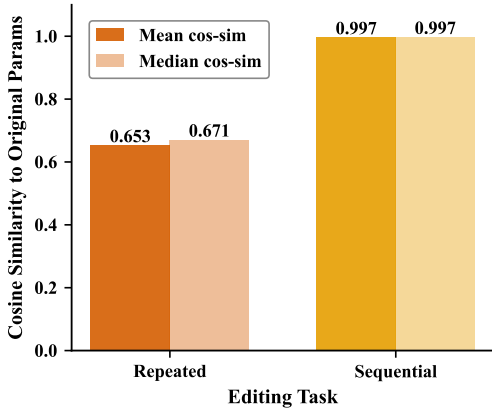
We repeat the repeated-editing experiment on GPT-2-



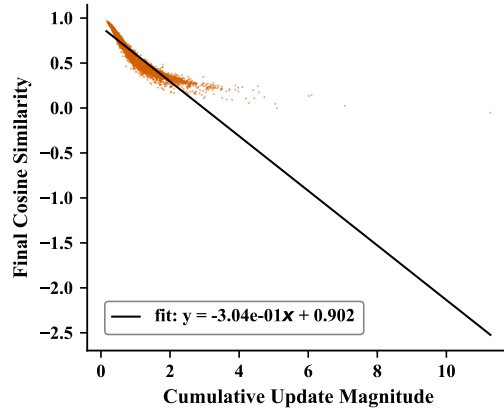
(a) Layer 6: Lorenz Curve



(b) Layer 6: Update Direction Similarity



(c) Layer 6: Parameter Deformation



(d) Layer 6: Mag. vs Cos-sim (Spearman $\rho = -0.990$)

Fig. 3. Mechanistic evidence on layer 6. (a) Lorenz curves of cumulative update magnitude (Gini coefficients annotated in the legend). (b) Average pairwise cosine similarity of the update vectors across edits. (c) Mean/median cosine similarity of the edited parameters vs. the original parameters. (d) Per-neuron scatter of cumulative update magnitude vs. final cosine similarity (Spearman ρ annotated).

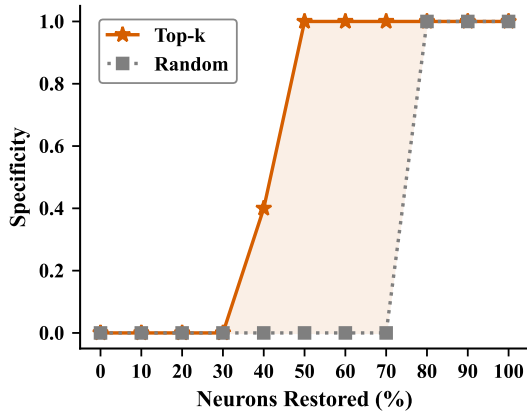


Fig. 4. Neuron-restoration experiment: Specificity vs. the fraction of neurons restored, either by top- k (largest cumulative update magnitude) or randomly.

XL [1]; the results are reported in Appendix D. The phenomenon is qualitatively similar but milder: repeated editing

also degrades Specificity and induces update concentration (higher Gini) on GPT-2-XL, but to a lesser extent, consistent with its milder performance degradation, indicating that the above conclusions are not limited to a single model.

IV. METHODOLOGY

In this section, we introduce our proposed method COOL-Edit in Section IV-A and provide a preliminary validation in Section IV-B.

A. COOL-Edit

The analysis in Section III-B traces the collapse under repeated editing back to the excessive updating of a small number of neurons. To alleviate this problem, we look for inspiration from existing methods. NMKE [21] is a classic improvement over AlphaEdit: by dynamically sparsifying the neuron mask and selecting “knowledge-general” and “knowledge-specific” neurons to edit, it performs well on lifelong editing tasks. Under repeated editing, Fig. 5 shows that NMKE clearly outperforms AlphaEdit.

We hypothesize that the benefit of NMKE may not come from its specific neuron-selection strategy, but from the masking mechanism itself—that only a subset of neurons is updated each time. Since the selected neurons differ across edits, the updates are spread more evenly across neurons, thereby alleviating neuron overheating. To test this hypothesis, we introduce a random-mask baseline (**Random**) that performs the same sparse editing as NMKE but selects the neurons to update completely at random. Fig. 5 shows that Random performs comparably to NMKE on Efficacy and Generalization, and is even clearly higher than NMKE on Specificity. In Fig. 6, compared with AlphaEdit, NMKE effectively reduces parameter deformation, and the random baseline reduces it even more. This further supports our hypothesis.

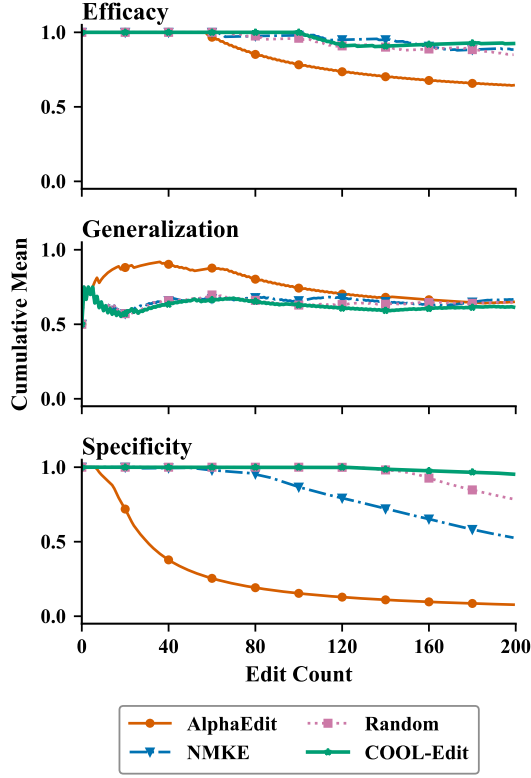


Fig. 5. Cumulative means of the three metrics over 200 repeated edits for the four methods.

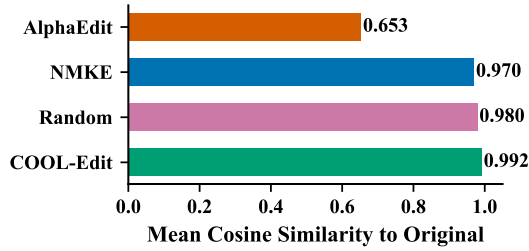


Fig. 6. Average cosine similarity of the four methods on layer 6 with respect to the original parameters (**higher means less deformation**).

The above analysis confirms the effectiveness of the sparse-

masking editing mechanism for alleviating neuron overheating. Accordingly, we propose **COOL-Edit** (Cold-neuron Optimized Overheating-Limited Knowledge Editing): on top of sparse-masked editing, we maintain a cumulative *heat* value for each neuron; the hotter a neuron is, the lower the probability that it is updated, so that overheating is suppressed in a more targeted way.

Specifically, COOL-Edit maintains a scalar heat $h_i^{(t)}$ for each neuron i in each edited layer l , initialized to 0. At editing step $t + 1$, the base editor (e.g., AlphaEdit) produces an unconstrained update vector $\Delta\tilde{W}_{l,i}^{(t+1)}$, and COOL-Edit decides which neurons actually receive the update:

- 1) **Softmax sampling probability.** The hotter a neuron is, the lower the probability that it is sampled (“cooling down”):

$$p_i^{\text{soft}} = \frac{\exp(-h_i^{(t)}/\tau)}{\sum_{j=1}^{n_l} \exp(-h_j^{(t)}/\tau)}, \quad (1)$$

where $\tau > 0$ is the softmax sampling temperature (a hyper-parameter, not to be confused with the heat $h_i^{(t)}$), and n_l is the number of neurons in layer l .

- 2) **Scaling to the expected number of edits.** We scale the softmax probability by ρn_l (where $0 < \rho < 1$ is the fraction of neurons to select, a hyper-parameter) and truncate it to 1, so that the expected number of updated neurons is close to ρn_l :

$$p_i = \min(1, \rho n_l \cdot p_i^{\text{soft}}) \quad (2)$$

- 3) **Bernoulli sampling.** Each neuron independently samples a binary mask $m_i \sim \text{Bernoulli}(p_i)$, and only the neurons selected by the mask receive the update:

$$\Delta W_{l,i}^{(t+1)} = m_i \cdot \Delta\tilde{W}_{l,i}^{(t+1)}. \quad (3)$$

- 4) **Heat update.** After applying the update, the heat is decayed by a factor $0 < \gamma < 1$ and accumulated with the actual update magnitude:

$$h_i^{(t+1)} = \gamma h_i^{(t)} + \|\Delta W_{l,i}^{(t+1)}\|_2. \quad (4)$$

The complete procedure is summarized in Algorithm 1.

Algorithm 1 COOL-Edit (one editing step)

Require: layer weights W_l , heat vector h_l , edit request (k, v) , hyper-parameters (τ, ρ, γ)

Ensure: updated weights W_l and heat h_l

- 1: Compute the unconstrained update $\Delta\tilde{W}$ neuron by neuron via the base editor (e.g., AlphaEdit)
 - 2: **for** $i = 1$ **to** n_l **do**
 - 3: $p_i^{\text{soft}} = \exp(-h_i/\tau) / \sum_j \exp(-h_j/\tau)$
 - 4: $p_i = \min(1, \rho n_l \cdot p_i^{\text{soft}})$
 - 5: sample $m_i \sim \text{Bernoulli}(p_i)$ independently
 - 6: $\Delta W_{l,i} = m_i \cdot \Delta\tilde{W}_{l,i}$
 - 7: $h_i = \gamma h_i + \|\Delta W_{l,i}\|_2$
 - 8: **end for**
 - 9: **return** W_l, h_l
-

B. Preliminary Validation on Repeated Editing

The previous section argued for the effectiveness of masked editing at the mechanism level and positioned COOL-Edit as an improvement that adds a heat constraint on top of sparse masking. This subsection compares the final performance of the four methods on the 200-edit repeated-editing task to validate the gains of COOL-Edit on the three metrics.

Among the three metrics in Fig. 5, COOL-Edit performs comparably to NMKE and the random baseline on Efficacy and Generalization, and clearly better than AlphaEdit on Efficacy; on Specificity, COOL-Edit stands out and is clearly better than all the other methods. In the parameter-deformation comparison of Fig. 6, COOL-Edit achieves the highest average cosine similarity on layer 6—the least deformation—indicating a stronger ability to alleviate neuron overheating. We also evaluate the model’s general capabilities after editing with MMLU [24]. In Fig. 7, COOL-Edit retains a generation accuracy of 0.426, clearly higher than all the other methods. These results consistently show that the heat constraint further suppresses parameter deformation on top of the masking mechanism, better preserves the model’s general capabilities, and avoids model collapse.

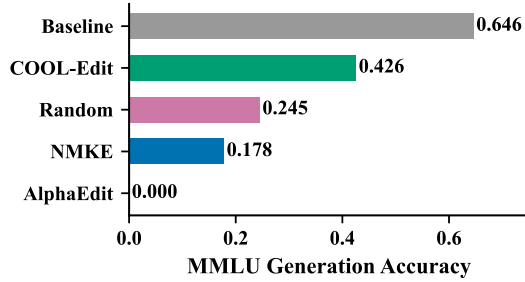


Fig. 7. MMLU generation accuracy of each method after 200 repeated edits (Baseline denotes the unedited model).

C. Complexity Analysis

The additional overhead of COOL-Edit over the base editor consists of one $O(n_i)$ softmax, n_i Bernoulli samples, and n_i scalar heat updates—negligible compared with the time and memory cost of the base editor.

V. EXPERIMENTS

In this section, we validate the effectiveness of our method on the serial lifelong knowledge editing task.

A. Experimental Setup

Benchmark. We perform *serial lifelong knowledge editing* on CounterFact-freq, a dataset generated from CounterFact [11] (see Appendix B), and evaluate the performance. Each piece of knowledge in this dataset is paired with 5 to 50 editing targets, simulating knowledge with different update frequencies in real-world scenarios; knowledge with higher update frequency is less numerous, consistent with the long-tail law of knowledge updates in real life [26]. To evaluate the editing performance, we adopt three main metrics (see

Appendix B): **Efficacy** (whether the edit takes effect on the original prompt), **Generalization** (whether it generalizes to paraphrased prompts), and **Specificity** (whether nearby facts are preserved). Each checkpoint is evaluated based on the knowledge state after the current edits. In addition, we use MMLU [24] to evaluate the general capabilities of the model.

Baselines. Our method COOL-Edit is built on top of AlphaEdit and is inspired by NMKE, so we take these two methods as baselines. We edit the three methods on LLaMA-3-8B-Instruct [2] and report the metrics at the checkpoints $T \in \{10, 100, 500, 1000, 1500, 2000\}$. More details are given in Appendix A.

B. Main Results

Table I lists Efficacy, Generalization, and Specificity of AlphaEdit, NMKE, and COOL-Edit at the checkpoints 10 / 100 / 500 / 1000 / 1500 / 2000. Although the metrics of COOL-Edit are relatively low when the number of edits is small, from $T = 500$ onward it clearly leads the other methods, and in particular is significantly better than AlphaEdit. This reflects the strong capability of COOL-Edit on the serial lifelong knowledge editing task.

Table II reports the average per-step time over the 2000 edits and the MMLU generation accuracy at $T = 1000$ and $T = 2000$ for the three methods. COOL-Edit is the fastest (12.13 s/step, vs. 14.80 s/step for AlphaEdit and 39.52 s/step for NMKE), and at $T = 1000$ it retains the highest MMLU general capability (0.188, vs. 0.001 for AlphaEdit and 0.004 for NMKE). By $T = 2000$, all three methods collapse completely (MMLU is 0.000 for all). This reflects the higher editing efficiency of COOL-Edit and its better protection of the model’s general capabilities.

It is worth noting that COOL-Edit is built on the complete AlphaEdit and adds a neuron-selection mechanism, yet it is even faster than AlphaEdit. We investigated this interesting phenomenon and found that the reason lies in the step where AlphaEdit solves for v by optimization. In this step, the optimizer stops when the loss falls below a threshold (0.05) or the number of optimization steps reaches an upper bound (25). The fewer final optimization steps, the less time is consumed. Fig. 8 plots the cumulative mean of the number of optimization steps per edit over the 2000 edits. The number of optimization steps of AlphaEdit first decreases and then rises (mean 22.65), whereas COOL-Edit and NMKE remain low overall after decreasing, despite a slight rebound (12.93 and 13.26). The smaller number of optimization steps lets COOL-Edit and NMKE save a great deal of time in this step, while the additional time overhead of COOL-Edit’s neuron-selection mechanism is very small, so its total time ends up being shorter than AlphaEdit’s; NMKE’s neuron-selection mechanism is very time-consuming, hence its total time is the longest. As for why the number of optimization steps behaves in this way, we conjecture that methods such as AlphaEdit drift the model parameters toward some direction favorable to optimization, so that the number of optimization steps gradually decreases, but after many edits the model’s neurons overheat and the

TABLE I

THE THREE MAIN METRICS AT EACH CHECKPOINT (T) ON THE 2000-EDIT SERIAL LIFELONG KNOWLEDGE EDITING BENCHMARK. “EFF.”, “GEN.”, AND “SPE.” DENOTE EFFICACY, GENERALIZATION, AND SPECIFICITY, RESPECTIVELY; $\uparrow$ INDICATES THAT HIGHER IS BETTER. THE BEST RESULT IN EACH COLUMN IS IN **BOLD** AND THE SECOND-BEST IS UNDERLINED.

Method	$T=10$			$T=100$			$T=500$			$T=1000$			$T=1500$			$T=2000$		
	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$	Eff. $\uparrow$	Gen. $\uparrow$	Spe. $\uparrow$
AlphaEdit	1.00	0.75	<u>0.58</u>	1.00	0.86	0.51	0.67	0.42	0.11	0.81	0.40	0.11	0.31	0.24	0.05	0.27	0.17	0.10
NMKE	1.00	0.75	<u>0.58</u>	1.00	<u>0.80</u>	<u>0.56</u>	0.98	<u>0.72</u>	<u>0.39</u>	<u>0.94</u>	<u>0.75</u>	<u>0.22</u>	<u>0.59</u>	0.49	<u>0.15</u>	<u>0.59</u>	<u>0.41</u>	<u>0.14</u>
COOL-Edit (Ours)	1.00	<u>0.69</u>	0.59	1.00	0.76	0.59	0.98	0.74	0.43	0.99	0.83	0.24	0.62	<u>0.41</u>	0.17	0.60	0.48	0.15

TABLE II

AVERAGE PER-STEP TIME AND MMLU GENERATION ACCURACY AT $T=1000$ AND $T=2000$ ON THE 2000-EDIT BENCHMARK. $\downarrow$ INDICATES THAT LOWER IS BETTER AND $\uparrow$ THAT HIGHER IS BETTER. THE BEST RESULT IN EACH COLUMN IS IN **BOLD** AND THE SECOND-BEST IS UNDERLINED.

Method	Avg. Step Time (s) $\downarrow$	MMLU $\uparrow$	
		$T=1000$	$T=2000$
AlphaEdit	<u>14.80</u>	0.001	0.000
NMKE	39.52	<u>0.004</u>	0.000
COOL-Edit (Ours)	12.13	0.188	0.000

parameters become severely deformed, causing the number of optimization steps to rise again. COOL-Edit and NMKE alleviate neuron overheating, so the rebound is smaller; COOL-Edit alleviates overheating more strongly, so its rebound is even weaker than NMKE’s, and from around $T = 1300$ onward its number of steps is stably lower than NMKE’s. However, we currently have no direct evidence for this conjecture, and the underlying mechanism warrants future study.

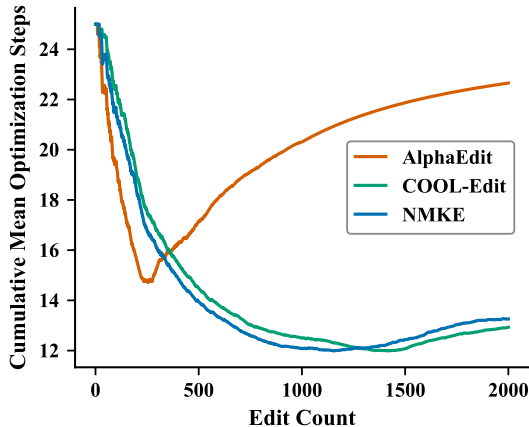


Fig. 8. Cumulative mean of the number of optimization steps per edit over the 2000 edits.

VI. CONCLUSION

In this paper, we study serial lifelong knowledge editing, in which the same fact must be updated repeatedly at high

frequency. Through a controlled repeated-editing analysis, we reveal a key failure mode of existing editors: repeatedly editing the same subject-relation pair concentrates the updates on a small, localized subset of neurons. We term this phenomenon neuron overheating and support it with three converging pieces of evidence: the cumulative update magnitude is highly concentrated, the deformed neurons show a dramatic drop in per-neuron cosine similarity, and cumulative update magnitude correlates strongly negatively with parameter deformation. A neuron-restoration experiment further confirms that these overheated neurons are the main cause of performance degradation and model collapse.

Based on these observations, we propose COOL-Edit, a heat-guided neuron selection mechanism built on top of sparse-masked editing. COOL-Edit maintains a cumulative heat value for each neuron and samples neurons with probability inversely proportional to their heat, so that successive edits are spread across the model and no single neuron is overused. On the CounterFact-freq benchmark over 2,000 edits, COOL-Edit clearly leads the baselines from the $T = 500$ checkpoint onward in Efficacy, Generalization, and Specificity, retains the highest MMLU accuracy among all methods, and is even faster than its base editor AlphaEdit. These results demonstrate that heat-aware neuron selection is an effective and efficient principle for lifelong knowledge maintenance of LLMs and a promising way to prevent catastrophic interference in continual model updating.

VII. FUTURE WORK

Our work suggests several promising directions. First, we observe that the number of optimization steps of AlphaEdit first decreases and then rebounds during repeated editing, and conjecture that the rebound stems from neuron overheating, but direct evidence is still lacking; studying this mechanism would deepen our understanding of how knowledge editing deforms model internals. Second, although COOL-Edit has been validated on LLaMA-3-8B, scaling it to larger models and more diverse architectures is needed to confirm its generality. Third, COOL-Edit currently builds on AlphaEdit as its base editor; combining the heat-aware selection mechanism with other editors, such as MEMIT or neuron-level methods, may further improve editing stability. Finally, extending heat-guided selection to hybrid settings that combine parameter-preserving approaches (e.g., retrieval-augmented generation)

with parameter-modifying editing, and to more complex knowledge forms such as multi-hop and structured knowledge, remains an interesting avenue for future research.

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REFERENCES

- [1] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," OpenAI, San Francisco, CA, USA, Tech. Rep., Feb. 2019.
- [2] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, et al., "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.
- [3] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., "GPT-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.
- [4] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, et al., "DeepSeek-V3 technical report," *arXiv preprint arXiv:2412.19437*, 2024.
- [5] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.
- [6] A. Lazaridou, A. Kuncoro, E. Gribovskaya, D. Agrawal, A. Liska, T. Terzi, et al., "Mind the gap: Assessing temporal generalization in neural language models," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 15995–16008.
- [7] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, et al., "Overcoming catastrophic forgetting in neural networks," *Proc. National Academy of Sciences of the United States of America*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [8] A. Gupta, A. Rao, and G. Anumanchipalli, "Model editing at scale leads to gradual and catastrophic forgetting," in *Findings of the Association for Computational Linguistics*, Bangkok, Thailand, Aug. 2024, pp. 15202–15232.
- [9] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, and H. Wang, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2312.10997*, 2023.
- [10] C. Zheng, L. Li, Q. Dong, Y. Fan, Z. Wu, J. Xu, and B. Chang, "Can we edit factual knowledge by in-context learning?," in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing*, Singapore, 2023, pp. 4862–4876.
- [11] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, "Locating and editing factual associations in GPT," in *Advances in Neural Information Processing Systems*, vol. 35, New Orleans, LA, USA, 2022, pp. 17359–17372.
- [12] J. Fang, H. Jiang, K. Wang, Y. Ma, J. Shi, X. Wang, X. He, and T.-S. Chua, "AlphaEdit: Null-space constrained knowledge editing for language models," in *The 13th International Conference on Learning Representations*, 2025.
- [13] B. Dhingra, J. R. R. Cole, J. M. Eisenschlos, D. Gillick, J. Eisenstein, and W. W. Cohen, "Time-aware language models as temporal knowledge bases," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 257–273, 2022.
- [14] Y. Cheng, Y.-C. Yu, K.-P. Chang, and Y.-C. F. Wang, "Serial lifelong editing via mixture of knowledge experts," in *Proc. 63rd Annual Meeting of the Association for Computational Linguistics*, Vienna, Austria, Jul. 2025, pp. 30888–30903.
- [15] Z. Zhong, Z. Wu, C. D. Manning, C. Potts, and D. Chen, "MQUAKE: Assessing knowledge editing in language models via multi-hop questions," in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing*, Singapore, Dec. 2023, pp. 15686–15702.
- [16] E. Mitchell, C. Lin, A. Bosselut, C. D. Manning, and C. Finn, "Memory-based model editing at scale," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, Baltimore, MD, USA, Jul. 2022, pp. 15817–15831.
- [17] E. Mitchell, C. Lin, A. Bosselut, C. Finn, and C. D. Manning, "Fast model editing at scale," in *The 10th International Conference on Learning Representations*, 2022.
- [18] Q. Liu, J. Zhang, O. Wu, M. Ng, and Y. Du, "AlphaEdit+: Model editing in the presence of conflicting and inconsistent knowledge," in *Findings of the Association for Computational Linguistics: ACL 2026*, San Diego, California, United States: Association for Computational Linguistics, Jul. 2026, pp. 14 809–14 835, doi: 10.18653/v1/2026.findings-acl.728.
- [19] M. Geva, R. Schuster, J. Berant, and O. Levy, "Transformer feed-forward layers are key-value memories," in *Proc. 2019 Conf. Empirical Methods in Natural Language Processing*, Punta Cana, Dominican Republic, Nov. 2021, pp. 5484–5495.
- [20] K. Meng, A. S. Sharma, A. J. Andonian, Y. Belinkov, and D. Bau, "Mass-editing memory in a transformer," in *The 11th International Conference on Learning Representations*, 2023.
- [21] J. Liu, J. Sun, S. Shen, C. Yang, and S. Wang, "Edit less, achieve more: Dynamic sparse neuron masking for lifelong knowledge editing in LLMs," in *Advances in Neural Information Processing Systems*, vol. 38, San Diego, CA, USA, 2025.
- [22] Y. Yao, P. Wang, B. Tian, S. Cheng, Z. Li, S. Deng, H. Chen, and N. Zhang, "Editing large language models: Problems, methods, and opportunities," in *Proc. 2023 Conf. Empirical Methods in Natural Language Processing*, Singapore, 2023, pp. 10222–10240.
- [23] H. Jiang, J. Fang, T. Zhang, B. Bi, A. Zhang, R. Wang, T. Liang, and X. Wang, "Neuron-level sequential editing for large language models," in *Proc. 63rd Annual Meeting of the Association for Computational Linguistics*, Vienna, Austria, Jul. 2025, pp. 16678–16702.
- [24] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," in *The 9th International Conference on Learning Representations*, 2021.
- [25] P. Wang, N. Zhang, B. Tian, Z. Xi, Y. Yao, Z. Xu, M. Wang, S. Mao, X. Wang, S. Cheng, K. Liu, Y. Ni, G. Zheng, and H. Chen, "EasyEdit: An easy-to-use knowledge editing framework for large language models," in *Proc. 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Bangkok, Thailand, Aug. 2024, pp. 82–93.
- [26] L. S. Buriol, C. Castillo, D. Donato, S. Leonardi, and S. Millozzi, "Temporal analysis of the Wikigraph," in *Proc. 2006 IEEE/WIC/ACM Int. Conf. Web Intelligence*, Hong Kong, China, 2006, pp. 45–51.

APPENDIX A EXPERIMENTAL SETUP

Model. The main experiments are conducted on LLaMA-3-8B-Instruct [2]. The edited layers are layers 4–8, and each edited layer has 14,336 neurons. For all models we use HuggingFace Transformers by default.

Hyper-parameters. Our code is built on the EasyEdit [25] framework. The hyper-parameters of AlphaEdit (both the original AlphaEdit and AlphaEdit used as the base editor in other methods) follow the default configuration of this framework. For NMKE, we use the default settings of the corresponding open-source code released with the original paper. For COOL-Edit, we set the softmax sampling temperature $\tau = 1$, the heat decay coefficient $\gamma = 0.9$, and the neuron selection ratio $\rho = 0.8$.

Hardware. All experiments are conducted on an NVIDIA A100 80 GB GPU.

Prompt for Dataset Generation

You are an expert in generating question-answer datasets. Given a prompt and an original answer about that prompt, you need to generate {NUM_ANSWERS} plausible alternative answers of the same type as the original answer. Here is the information:

- Prompt: {PROMPT_TEMPLATE}
- Original answer: {ORIGINAL_ANSWER}

RULES:

- Each answer must be a SINGLE WORD (no spaces, no hyphens, no parentheses, no punctuation)
- Do NOT use compound words connected by hyphens (e.g., "German-American")
- Do NOT use abbreviations followed by full forms (e.g., "USA (United States)")
- Each answer must be of the same semantic type as the original answer

Please output ONLY the {NUM_ANSWERS} alternative answers, each on a separate line, without any numbering, bullet points, or additional text.

Fig. A.1. Prompt to use deepseek-v4-flash to generate dataset of CounterFact-freq based on CounterFact

APPENDIX B DATASET DETAILS

A. CounterFact-freq

CounterFact-freq is generated from the standard CounterFact knowledge-editing records. For each record, we retain its original prompt, subject and ground truth. Only the target object (o^*) undergoes sequential modifications to simulate the temporal evolution of real-world facts.

Serial Lifelong Knowledge Editing (sLKE) demands updating the same concept multiple times. To facilitate this, we prompt DeepSeek-V4-flash to generate multiple plausible, distinct, and semantically consistent alternative target objects for each subject-relation pair. The prompt we used is shown in Fig. A.1.

To replicate the dynamics of real-world knowledge shifts, we select 138 unique records and assign them varying update frequencies across 10 gradient levels. All generated edit sequences are randomly shuffled with a fixed random seed. This produces a single, continuous 2000-step editing schedule.

B. Evaluation Metrics

Each piece of knowledge i corresponds to a fact triple (s_i, r_i, o_i) , where o_i is the original target (target_true) and is accompanied by a set of candidate editing targets (target_new) $\{o_i^{(1)}, \dots, o_i^{(K)}\}$. In our editing task, the target value of the same fact switches among the candidate targets; we denote the target of the current editing step (target_now) by $o_i^{\text{now}} \in \{o_i^{(1)}, \dots, o_i^{(K)}\}$. The union of all targets forms the set $\mathcal{O}_i = \{o_i\} \cup \{o_i^{(1)}, \dots, o_i^{(K)}\}$. Let $p(s_i, r_i)$ be the original prompt and $P_G[o | p]$ the probability that model G generates the object o given prompt p . All three main metrics require the probability of the designated target (the current editing target o_i^{now} or the original target o_i) to be *greater than that of all the remaining targets in the set*. This “comparison against all targets” design differs from CounterFact [11], which compares only against the single original target, and better fits the multi-target editing setting. Each metric takes the expectation E_i over records.

Efficacy (rewrite_success): on the original prompt, the edited model should make the current editing target dominate all other targets in the set (including the original target and the other candidates):

$$\text{Efficacy} = E_i [P_G[o_i^{\text{now}} | p(s_i, r_i)] > \max_{o \in \mathcal{O}_i \setminus \{o_i^{\text{now}}\}} P_G[o | p(s_i, r_i)]] \quad (\text{A.1})$$

Generalization (paraphrase_success): the same requirement extended to paraphrased prompts:

$$\text{Generalization} = E_i [E_{p \in \text{paraphrases}(s_i, r_i)} [P_G[o_i^{\text{now}} | p]] > \max_{o \in \mathcal{O}_i \setminus \{o_i^{\text{now}}\}} P_G[o | p]] \quad (\text{A.2})$$

Specificity (neighborhood_success): on neighborhood prompts, the model should keep the original target—i.e., the original target should dominate all other targets in the set (including the current editing target):

$$\text{Specificity} = E_i [E_{p \in \text{neighborhood}(s_i, r_i)} [P_G[o_i | p]] > \max_{o \in \mathcal{O}_i \setminus \{o_i\}} P_G[o | p]] \quad (\text{A.3})$$

TABLE A.1
PER-LAYER AVERAGE PAIRWISE COSINE SIMILARITY OF THE UPDATE VECTORS (REP = REPEATED EDITING, SEQ = SEQUENTIAL EDITING).

Layer	Pairwise cos-sim _{Rep}	Pairwise cos-sim _{Seq}
4	0.873	0.405
5	0.616	0.429
6	0.564	0.463
7	0.550	0.484
8	0.542	0.486

TABLE A.2
PER-LAYER PARAMETER DEFORMATION STATISTICS (REP = REPEATED EDITING, SEQ = SEQUENTIAL EDITING).

Layer	Mean $\cos\text{-sim}_{\text{Rep}}$	Median $\cos\text{-sim}_{\text{Rep}}$	Mean $\cos\text{-sim}_{\text{Seq}}$	Median $\cos\text{-sim}_{\text{Seq}}$
4	0.985	0.996	0.998	0.998
5	0.716	0.727	0.997	0.998
6	0.653	0.671	0.997	0.997
7	0.535	0.538	0.994	0.996
8	0.352	0.338	0.989	0.992

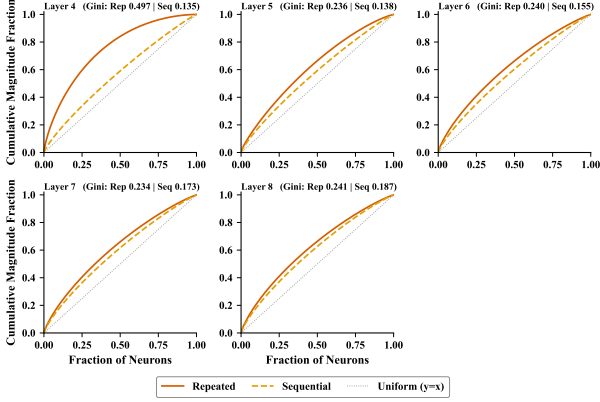


Fig. A.2. Lorenz curves of the cumulative update magnitude over the five layers (4–8): repeated editing vs. sequential editing.

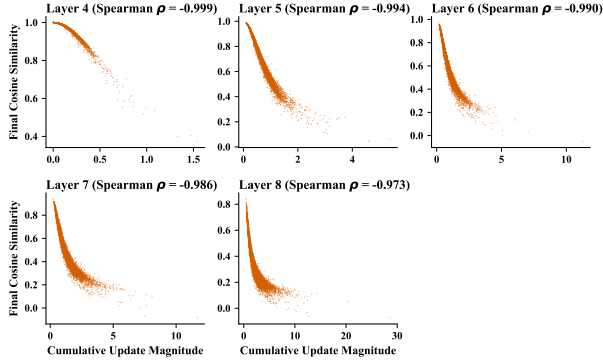


Fig. A.3. Per-neuron scatter of cumulative update magnitude vs. final cosine similarity over the five layers (4–8), with the Spearman ρ annotated.

APPENDIX C FULL RESULTS ACROSS EDITED LAYERS

This section reports the complete data over all five edited layers (4–8), corresponding to the subplots of Fig. 3 in the main text: Table A.1 reports the per-layer pairwise average similarity (corresponding to Fig. 3(b), supporting Observation 1); Table A.2 reports the per-layer parameter-deformation statistics (corresponding to Fig. 3(c), supporting Observation 2); Fig. A.2 shows the Lorenz curves of cumulative update magnitude (corresponding to Fig. 3(a), supporting Observation 1); Fig. A.3 shows the per-neuron scatter of cumulative update magnitude vs. final cosine similarity with the Spearman ρ annotated (corresponding to Fig. 3(d), supporting Observation 3).

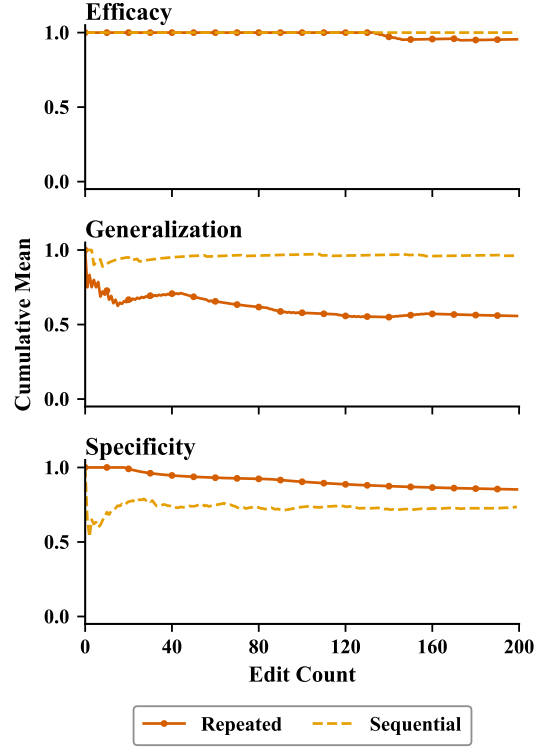


Fig. A.4. Cumulative means of the three metrics for repeated editing vs. sequential editing on GPT-2-XL.

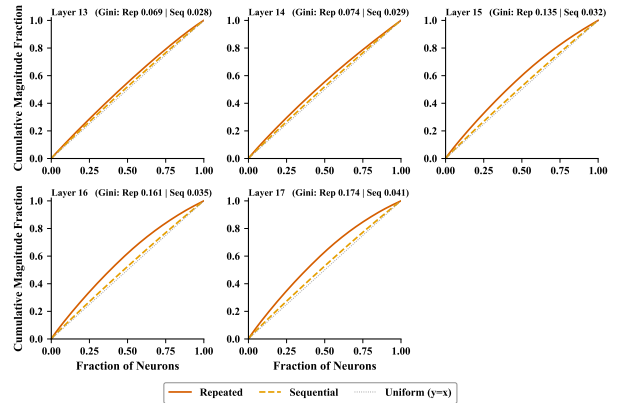


Fig. A.5. Lorenz curves of the cumulative update magnitude over the five layers (13–17) on GPT-2-XL: repeated editing vs. sequential editing.

APPENDIX D RESULTS ON GPT-2-XL

Fig. A.4 shows the cumulative means of the three metrics for repeated editing (Repeated) vs. sequential editing (Sequential) on GPT-2-XL; Fig. A.5 shows the Lorenz curves for layers 13–17 (the Gini coefficient is annotated in each title). The update-concentration phenomenon also exists on GPT-2-XL but is milder, supporting the cross-model consistency validation of Section III-B.